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Interactive Dashboards with Plotly Dash

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1. Executive Summary

This document presents the implementation of two interactive web dashboards for the **Real Madrid Pulse** data engineering project, developed as part of Workshop 4 of the Data Analysis Programming course. The dashboards constitute the final user-facing layer of the Medallion Architecture pipeline (Bronze → Silver → Gold), reading exclusively from Gold Parquet files generated by the PySpark Gold DAG.

Two independent Plotly Dash applications were developed: a **Governance Dashboard** running at `localhost:8052`, designed for technical audiences to monitor pipeline health and data quality KPIs; and a **Storytelling Dashboard** running at `localhost:8051`, designed for the functional user to explore sentiment trends, keyword insights, and source comparisons about Real Madrid public opinion.

The full pipeline was verified end-to-end, processing **280 records** (200 from web scraping, 80 from Reddit) with 100% schema compliance. The storytelling dashboard is powered by eleven Gold aggregation Parquet files `sentiment_distribution`, `sentiment_trend`, `top_keywords`, `top_bigrams`, `keyword_sentiment`, `source_comparison`, `volume_trend`, `aspect_sentiment`, `aspect_cowords`, `aspect_snippets`, and `sentiment_dist_time` and presents a cohesive narrative centered on three analytical questions: the overall sentiment polarity, the dominant discussion topics, and the contrast between journalistic and fan perspectives. A global positive sentiment rate of **56.43%** with an average VADER compound score of **0.2068** is the central finding communicated to the functional user.

2. End-to-End Integration Verification

2.1 Pipeline Execution Verification

Prior to dashboard development, the complete pipeline was executed from a clean state to confirm that all layers produce consistent and traceable outputs. The execution sequence followed was:

1. **Bronze ingestion DAG** (`football_bronze_pipeline`, `reddit_bronze_pipeline`): Raw data captured from `football-espana.net` and `r/realmadrid` in JSON/CSV format.
2. **Silver transformation DAG** (`football_silver_pipeline`, `reddit_api_silver_pipeline`): Schema enforcement, deduplication, text cleaning, IQR outlier capping, and dual persistence into PostgreSQL and Parquet.
3. **Gold PySpark DAG** (`gold_processing_dag`): Full NLP pipeline with spaCy and VADER, governance KPI computation, and seven storytelling aggregation files.

All DAGs completed with `success` status in the Airflow UI. The Gold Parquet files were confirmed present in `datalake_gold/` with the expected schemas and record counts. Both dashboard applications were started independently and loaded data without errors.

2.2 Integration Adjustments Made in Workshop 4

The following adjustments were applied specifically in this workshop to ensure correct dashboard-to-Gold connectivity:

- **Parquet file discovery:** Both dashboard applications use `glob.glob()` with a wildcard pattern (`prefix_*.parquet`) and `max(files, key=os.path.getmtime)` to always load the most recent Gold file for each aggregation prefix. This prevents stale data from being displayed after a new Gold DAG run.
- **Empty DataFrame handling:** All chart callbacks include explicit guards (`if not df.empty`) to render empty figures gracefully when Gold files are absent, preventing dashboard crashes during cold starts.
- **Port configuration:** The storytelling dashboard was confirmed to run at `localhost:8051` and the governance dashboard at `localhost:8050`, maintaining application separation as required by the workshop specification.
- **Auto-refresh interval:** A `dcc.Interval` component with a 5-minute refresh cycle was added to both dashboards, ensuring that a new Gold DAG run is reflected in the UI without requiring a manual browser reload.

3. Governance Dashboard

3.1 Layout Description and Design Rationale

The Governance Dashboard was designed for a technical audience responsible for monitoring data quality, completeness, operational compliance, and ingestion performance of the Real Madrid Pulse pipeline. The interface follows a top-to-bottom monitoring workflow in which users first observe high-level indicators and then progressively explore more detailed quality metrics.

The dashboard begins with a header section containing the dashboard title, project identification information, team members, and the timestamp of the latest available Gold layer update. This section provides immediate context regarding the current execution state of the pipeline.

Immediately below the header, a filter bar allows users to dynamically change the dashboard scope. The filter bar contains three controls:

- A **source dropdown** with the options *All Sources*, *Reddit*, and *Web Scraping*.
- A **period selector** implemented as radio buttons (Daily, Weekly, Monthly) that modifies the aggregation level of temporal analyses.
- A **Refresh Latest Gold** button that forces the dashboard to reload the most recent Gold Parquet files.

The first analytical section consists of four KPI summary cards arranged horizontally. These cards provide an immediate overview of the most important governance indicators:

1. Total Records.
2. Body Text Missing Rate.
3. Duplicate Rate.
4. Schema Compliance Rate.

Below the KPI cards, the dashboard presents the **Null Rate by Key Field** chart. This horizontal bar chart ranks critical fields according to their percentage of missing values, allowing rapid identification of completeness problems.

The next row contains two complementary visualizations. On the left side, the **Volume of Records by Period and Source** chart displays the number of records ingested during each period, grouped by source (Reddit and Web Scraping). This visualization allows users to identify changes in collection activity and compare source contributions over time. On the right side, the **Outlier Rate by Numeric Field** chart summarizes anomaly rates for variables such as reading time, Reddit score, and number of comments.

The dashboard also includes a dedicated section for **Ingestion Compliance Monitoring**. These indicators compare the expected number of collection executions with the actual number of successful pipeline runs for Reddit and Web Scraping sources. This component allows users to evaluate operational reliability and detect potential collection interruptions.

The final section contains the **Governance KPI Table**. This component displays all calculated governance metrics grouped by category and supports sorting and filtering. The table serves as the detailed operational view after users have inspected the higher-level visual summaries.

This layout follows the principle of "overview first, details on demand", allowing users to identify quality issues rapidly before investigating specific metrics in greater depth.

3.2 Component List and KPI Mapping

Dashboard Component	Governance KPI(s)	Gold Source File
KPI Card: Total Records	total_records	governance_*.parquet
KPI Card: Body Text Missing	body_text_missing_rate	governance_*.parquet
KPI Card: Duplicate Rate	duplicate_rate	governance_*.parquet
KPI Card: Schema Compliance	schema_compliance_rate	governance_*.parquet
Source Filter	records_reddit, records_scraping and dynamically recalculated metrics	gold_realmadrid_*.parquet
Period Selector	Temporal aggregation of record volume	gold_realmadrid_*.parquet
Refresh Button	Latest governance snapshot reload	governance_*.parquet and gold_realmadrid_*.parquet

Table 1. Governance dashboard KPI cards and interactive controls.

Dashboard Component	Governance KPI(s)	Gold Source File
Null Rate Chart	null_rate_* metrics for key fields	governance_*.parquet
Volume Over Time Chart	total_records, records_reddit, records_scraping	gold_realmadrid_*.parquet
Outlier Rate Chart	outlier_rate_reading_time_min, outlier_rate_score, outlier_rate_num_comments	governance_*.parquet
Ingestion Compliance Metrics	ingestion_compliance_scraping, ingestion_compliance_reddit, actual_runs_scraping, actual_runs_reddit	governance_*.parquet
Language Distribution Metrics	pct_likely_spanish, pct_likely_english	governance_*.parquet
Text Length Statistics	text_length_mean, text_length_median, text_length_max, text_length_min	governance_*.parquet
Governance KPI Table	All governance KPIs including completeness, quality, ingestion, compliance, language and operational indicators	governance_*.parquet

Table 2. Governance dashboard visualizations and KPI mappings.

3.3 Design Rationale

The Governance Dashboard adopts a dark-theme visual design optimized for prolonged monitoring sessions and high information density. The color palette is based on three principal colors:

- **#0f172a**: Main dashboard background.
- **#111827**: Card and chart background surfaces.
- **#f1c40f**: Accent color used for highlights, titles, KPI values, and interactive elements.

Additional semantic colors were incorporated to communicate data quality status consistently across the dashboard:

- Green (#2ecc71) for healthy or compliant metrics.
- Red (#e74c3c) for critical quality issues.
- Yellow (#f1c40f) for warnings or moderate-risk conditions.
- Gray (#95a5a6) for neutral information and secondary indicators.

Several design decisions were made to improve usability and readability:

- KPI cards use large typography and centered alignment to maximize metric visibility.

- Charts are placed inside independent containers with rounded corners and subtle shadows, visually separating analytical sections.
- Consistent annotations indicate the data source and current filter scope for every visualization.
- Interactive controls are grouped into a dedicated filter bar to centralize dashboard navigation.
- The DataTable uses category-based color accents to distinguish Core, Completeness, Quality, Ingestion, and Compliance metrics.
- Plotly interactive features such as hover tooltips, zooming, and responsive resizing were enabled to support exploratory analysis.

Overall, the dashboard prioritizes rapid operational monitoring, allowing technical users to identify completeness problems, anomalous values, ingestion issues, schema violations, and compliance deviations with minimal cognitive effort.

4. Storytelling Dashboard

4.1 Dashboard Narrative and User Story

The storytelling dashboard (`dashboard/storytelling_app.py`) is designed for the functional user: a Real Madrid fan or sports analyst who wants to understand how the club is discussed across different media. The interface avoids technical jargon and prioritizes plain-language insight over raw numbers.

The narrative is structured around three analytical questions that guide the layout from top (overview) to bottom (detail):

1. **How positive or negative is the overall discourse?** Answered by the KPI cards, narrative summary card, and sentiment donut chart at the top of the page.
2. **Which topics drive the conversation and when?** Answered by the sentiment trend, volume activity, and top keywords components in the middle section.
3. **How do journalists and fans differ in their perspectives?** Answered by the source comparison chart and aspect-based sentiment chart in the lower section.

4.2 Layout Description

The application is organized as a single scrollable page with a fixed header and four visual rows:

1. **Header bar**: Project title ("Real Madrid Pulse"), subtitle ("Public Opinion & Sentiment Analysis Dashboard"), data sources (`Reddit r/realmadrid + Football-España`), last-updated timestamp, and team name. A gold accent border (`#f0a500`) is applied at the bottom of the header for brand consistency.

2. **KPI Cards row**: Four metric cards side by side — Total Mentions Analyzed, Dominant Sentiment (with percentage and label), Average Compound Score, and Most Active Source.

3. **Narrative Summary Card**: A single paragraph auto-generated from live Gold data, communicating the key insight in plain language for the functional user. It reports the dominant sentiment, positive/negative split, and the best and worst-scoring aspect by compound score.

4. **Sentiment & Activity Over Time row**: A Daily/Weekly toggle (`dbc.RadioItems`) controls two side-by-side charts: a line chart of average compound score per time period (left, 7/12 width) and a stacked bar chart of record volume per time period (right, 5/12 width).

5. **Distribution & Source Comparison row**: A donut chart of global sentiment distribution (left, 5/12 width) and a grouped bar chart comparing Reddit vs. Football-España across positive, neutral, and negative percentages (right, 7/12 width).

6. **Keywords & Aspect Sentiment row**: A horizontal bar chart of the top 15 keywords colored by sentiment mode (left, 5/12 width) and a horizontal bar chart of mean VADER compound per aspect with a zero-reference vertical line (right, 7/12 width).

4.3 Data Loading Architecture

All data is loaded exclusively from Gold Parquet files using `pd.read_parquet()`. The central loading function `load_latest_parquet(prefix)` discovers files via `glob.glob()` and selects the most recent one by modification time:

```
def load_latest_parquet(prefix: str) -> pd.DataFrame:
    pattern = os.path.join(GOLD_PATH, f"{prefix}_*.parquet")
    files   = glob.glob(pattern)
    if not files:
        return pd.DataFrame()
    latest = max(files, key=os.path.getmtime)
    try:
        return pd.read_parquet(latest)
    except Exception as e:
        print(f"ERROR␣reading␣{latest}:␣{e}")
        return pd.DataFrame()
```

Listing 1. Core data loading function — storytelling_app.py

A dedicated `load_full_gold()` function loads the complete Gold base dataset and derives temporal columns (`day` and `week`) from `published_at` at runtime, enabling the daily/weekly toggle without requiring separate pre-aggregated files for each granularity.

4.4 Color Palette and Visualization Standards

The dashboard applies a consistent dark-theme color palette defined in the `COLORS` dictionary at the top of the application. Sentiment encoding is standardized across all charts:

Table 3. Color Palette — Storytelling Dashboard

Role	Hex Code	Usage
Positive sentiment	`#2ecc71`	Donut, keywords, aspect bars, KPI cards
Neutral sentiment	`#95a5a6`	Donut, keywords, source comparison
Negative sentiment	`#e74c3c`	Donut, keywords, aspect bars
Reddit source	`#FF4500`	Trend lines, volume bars, source comparison
Scraping source	`#1a5276`	Trend lines, volume bars, source comparison
Accent (Real Madrid gold)	`#f0a500`	Header border, KPI values, narrative card border
Page background	`#0d1117`	Full page background
Card background	`#161b22`	All chart and KPI card containers

All charts include descriptive titles, labeled axes with units, legends, and hover tooltips. The aspect sentiment chart includes a vertical reference line at $x = 0$ to clearly separate positive from negative territory. The sentiment trend chart includes a horizontal reference line at $y = 0$ annotated as "Neutral". The y-axis of the sentiment trend is fixed to $[-1, 1]$ to maintain visual consistency across refresh cycles.

4.5 Callback Architecture

The application uses Plotly Dash callbacks triggered by a `dcc.Interval` component (5-minute auto-refresh cycle). Each chart is an independent callback, so a missing or corrupt Parquet file for one aggregation does not prevent other charts from rendering.

Table 4. Dashboard Callbacks — Outputs and Triggers

Callback	Output	Inputs
`update_timestamp`	`last-updated` text	`dcc.Interval`
`update_kpis`	KPI cards row + narrative card	`dcc.Interval`
`update_trend_charts`	Sentiment trend + volume chart	Granularity toggle + `dcc.Interval`
`update_donut`	Sentiment donut chart	`dcc.Interval`
`update_source_comparison`	Source comparison bar chart	`dcc.Interval`
`update_keywords`	Top keywords bar chart	`dcc.Interval`
`update_aspect_sentiment`	Aspect sentiment bar chart	`dcc.Interval`

4.6 KPI Cards and Narrative Summary Card

The `update_kpis` callback produces the four metric cards and the narrative card in a single execution to minimize redundant Parquet reads. The cards display:

- **Total Mentions Analyzed**: Sum of all record counts from the `sentiment_distribution` aggregation (280 in the last run).
- **Dominant Sentiment**: The label with the highest count ("positive", 56.4%), rendered in the corresponding sentiment color.
- **Average Compound Score**: Mean VADER compound across all Gold records (+0.207), colored green if positive or red if negative.
- **Most Active Source**: Source with the highest record count from `source_comparison` (Football-España, 200 records).

The narrative card generates a plain-language paragraph using the live Gold values, including the dominant sentiment percentage, positive/negative split, and the best and worst aspect by compound score:

```
narrative_text = (
    f"Analysis of {total:,} mentions from Reddit and Football-Espana shows "
    f"that public opinion about Real Madrid is {dominant.upper()} "
    f"({dominant_pct}% of all content, compound score: {compound_avg:+.3f}). "
    f"Positive sentiment accounts for {pos_pct}% of mentions, while negative "
    f"sentiment represents {neg_pct}%. The topic generating the most positive "
```

```
    f"reactions is {best_aspect}, while {worst_aspect} drives the most "
    f"critical discussion."
)
```

Listing 2. Narrative card text generation — update_kpis callback

4.7 Sentiment & Activity Over Time — Toggle Feature

The time series section implements a daily/weekly granularity toggle using `dbc.RadioItems`. Rather than reading pre-aggregated files, the `update_trend_charts` callback reads the full Gold base dataset and groups by the `day` or `week` column at runtime. This supports both granularities without requiring the Gold DAG to produce additional output files.

The sentiment trend chart plots one line per source (Reddit in orange `#FF4500`, Football-España in dark blue `#1a5276`), with the y-axis range fixed at $[-1, 1]$. The volume chart uses stacked bars per source with `barmode="stack"`, highlighting the contribution of each ingestion channel to total weekly activity.

4.8 Sentiment Donut Chart

The donut chart aggregates the `sentiment_distribution` Parquet file at the global level (summing counts across all sources) and renders a `go.Pie` trace with `hole=0.55`. A centered annotation inside the donut hole displays the dominant sentiment label and total record count. The legend is positioned horizontally below the chart to avoid overlap with the annotation.

4.9 Source Comparison Chart

The grouped bar chart renders three sentiment percentages (positive, neutral, negative) side by side for each source, using the `source_comparison` Parquet file with `barmode="group"`. Source keys (`"reddit"`, `"scraping"`) are mapped to display labels ("Reddit", "Football-España") at render time. The chart confirms that Football-España leads in positive sentiment (62%) while Reddit shows substantially higher neutrality (26.25%), consistent with the difference between sports journalism and discussion-oriented posts.

4.10 Top Keywords Bar Chart

The horizontal bar chart loads the top 15 keywords from `top_keywords` and enriches each entry with its `sentiment_mode` from `keyword_sentiment` via a left merge on the `keyword` field. Bar color reflects the dominant sentiment of each keyword (green/grey/red). Keywords not present in the sentiment file are assigned neutral color as a fallback. Keywords are sorted ascending by frequency so the highest-frequency term appears at the top of the horizontal chart.

4.11 Aspect Sentiment Bar Chart

The chart loads `aspect_sentiment` and filters to rows where `source == "all"`, displaying the global compound mean per aspect. Bars are colored green (≥ 0.05), red (≤ -0.05), or grey (between thresholds). A vertical reference line at $x = 0$ separates positive from negative territory, and the compound value is annotated outside each bar. The x-axis is fixed to $[-1, 1]$. The hover tooltip includes the mention count per aspect. The chart covers the 13 aspects in the expanded list: Mourinho, Mbappe, Vinicius, Carvajal, Transfers, Injuries, Arbeloa, Bellingham, Valverde, Barcelona, Clásico, Tchouameni, and Ancelotti.

4.12 Section 06 — Protagonist Deep Dive (*Why?*)

The sixth and most analytically detailed section of the storytelling dashboard addresses the explanatory question that the preceding sections leave open: given that a player or topic has a particular sentiment score, *why* does it have that score? What specific language drives the positive or negative classification?

The section is anchored by a topic selector implemented as a `dbc.RadioItems` pill group. The available topics correspond to the aspect list defined in the Gold DAG: Mourinho, Mbappé, Vinicius Jr., Carvajal, Transfers, Injuries, Arbeloa, Bellingham, Valverde, Barcelona, El Clásico, Tchouaméni, and Ancelotti. Selecting a topic triggers the `update_drilldown` callback, which loads three Gold Parquet files simultaneously:

- `aspect_cowords_*.parquet`: contains the most frequent co-occurring words for each aspect, split by sentiment context (positive vs. negative mentions). This file is produced by the Gold DAG by filtering the full corpus to records where the aspect appears, then counting token frequencies separately in positive and negative subsets.
- `aspect_snippets_*.parquet`: contains the representative text excerpts with the highest and lowest VADER compound scores for each aspect, one row per record. These snippets are the actual `body_text` or `title` values from the Gold base dataset, selected by ranking compound scores within the aspect-filtered subset.
- `aspect_sentiment_*.parquet`: provides the summary compound score, mention count, and positive/negative percentages for the selected aspect, displayed in a summary header above the three columns.

The layout presents three columns side by side. The left column shows a horizontal bar chart of the most frequent words co-occurring with the selected topic in **positive** contexts, colored green. The center column shows the equivalent chart for **negative** contexts, colored red. The right column displays representative text excerpts from both Reddit and Football-España, ordered from most positive to most negative compound score, with the source and score annotated for each excerpt.

This component bridges the quantitative sentiment metrics of Sections 01–05 with qualitative textual evidence, allowing the functional user to understand not only the score but the specific discourse that produces it. For example, selecting Tchouaméni reveals that his negative mentions co-occur with words such as *fight*, *training*, *incident*, and *hospital*, directly tracing the sentiment signal to the reported altercation with Valverde. Selecting Bellingham shows co-occurrence with *goal*, *brilliant*, and *season* in positive contexts, confirming that his score reflects genuine performance praise rather than sarcastic usage.

This section represents the most technically complex component of the dashboard, requiring three simultaneous Parquet reads, a dynamic topic filter, two bar charts, and a rendered text panel, all updated within a single callback triggered by the topic selector.

Component	**Description**	**Gold Source File**
Topic selector (pills)	Selects the active aspect for drilldown	—
Summary header	Compound score, mention count, pos/neg %	`aspect_sentiment_*.parquet`
Co-words positive chart	Top words in positive mentions of the aspect	`aspect_cowords_*.parquet`
Co-words negative chart	Top words in negative mentions of the aspect	`aspect_cowords_*.parquet`
Representative texts panel	Highest and lowest compound excerpts with source and score	`aspect_snippets_*.parquet`

Table 5. Section 06 components and Gold source files — Protagonist Deep Dive.

4.13 User Story Mapping

Table 6. User Story Mapping — Storytelling Dashboard

User Need	**Dashboard Component**	**Gold Aggregation**
Overall sentiment at a glance	KPI Cards + Donut Chart	`sentiment_distribution`
Sentiment evolution over time	Trend Line Chart	`gold_realmadrid` (live groupby)
Most discussed topics	Top Keywords Bar Chart	`top_keywords` + `keyword_sentiment`
Journalists vs. fans contrast	Source Comparison Bar Chart	`source_comparison`
Sentiment per player/topic	Aspect Sentiment Bar Chart	`aspect_sentiment`
Single key takeaway in plain language	Narrative Summary Card	Multiple files (computed live)
Volume peaks of public discussion	Volume Activity Bar Chart	`gold_realmadrid` (live groupby)

5. Visualization Design Decisions

5.1 Chart Type Justification

- **Donut chart for sentiment distribution:** With only three categories, a donut chart is appropriate and provides an immediate visual proportion. The hole allows a centered annotation with quantitative context that a pie chart cannot accommodate cleanly.
- **Horizontal bar charts for keywords and aspect sentiment:** Keyword labels and aspect names are long strings; horizontal orientation prevents label truncation. Sorting by value communicates ranking naturally, and the leftmost bar at zero provides a clear reference point for the aspect chart.
- **Line chart for sentiment trend:** Continuous temporal data over ordered time periods is most legible as a connected line. Multiple sources are overlaid as separate traces with distinct colors, enabling direct temporal comparison without additional chart area.
- **Stacked bar chart for volume:** Volume is an additive quantity where source contribution is relevant. Stacked bars display both total volume and per-source breakdown in a single chart. Bar charts are preferred over line charts here because volume is discrete (count per period), not a continuous quantity.
- **Grouped bar chart for source comparison:** With two sources and three sentiment percentages each, grouped bars allow direct side-by-side comparison per sentiment category. A stacked chart would obscure the individual percentages within each source.

5.2 Functional User Accessibility

All chart titles and axis labels use plain English without technical abbreviations. The narrative card provides a complete plain-language paragraph that functions as a standalone insight for users who do not engage with the charts. Sentiment colors (green/grey/red) follow universal convention and are applied consistently across all components, so the user learns the encoding once and applies it throughout the dashboard without repeated legend consultations.

5.3 Layout and Responsiveness

The dashboard is designed for standard 1080p resolution without horizontal scrolling. All chart heights are fixed between 300 and 380 pixels to maintain proportional layout. The Bootstrap 12-column grid system is used for all multi-column rows, with 5/7 column splits chosen to give more visual space to the richer or wider chart in each pair.

6. Results

6.1 Airflow UI — DAG Run History

Figure 1 presents the Apache Airflow user interface used to orchestrate the Real Madrid Pulse data pipeline. The screenshot shows the complete set of DAGs that compose the Bronze, Silver, and Gold layers of the architecture.

The Bronze layer is represented by the `football_scraping_universal_bronze` and `reddit_api_bronze_ingestion` DAGs, which are responsible for collecting data from web scraping sources and the Reddit API, respectively. The Silver layer is implemented through the `football_silver_pipeline` and `reddit_api_silver_pipeline` DAGs, where data cleaning, standardization, and quality validation processes are executed. Finally, the Gold layer is managed by the `gold_processing_dag`, which performs data integration, NLP processing, sentiment analysis, governance KPI generation, and storytelling aggregations.

The Airflow interface confirms that all DAGs are currently active and have been executed successfully. The *Last Run* column shows recent successful executions, while the *Recent Tasks* indicators display green status markers corresponding to completed tasks. Some failed executions are visible in the historical run counters; however, these failures correspond to earlier development and testing stages of the project, when DAG configurations and processing logic were still being validated. After the final implementation and debugging process, the pipeline reached a stable state and all production DAGs execute correctly.

This execution history demonstrates the operational readiness of the ETL architecture and verifies that data can flow automatically from the Bronze layer through Silver transformations and into the Gold analytical layer without manual intervention.

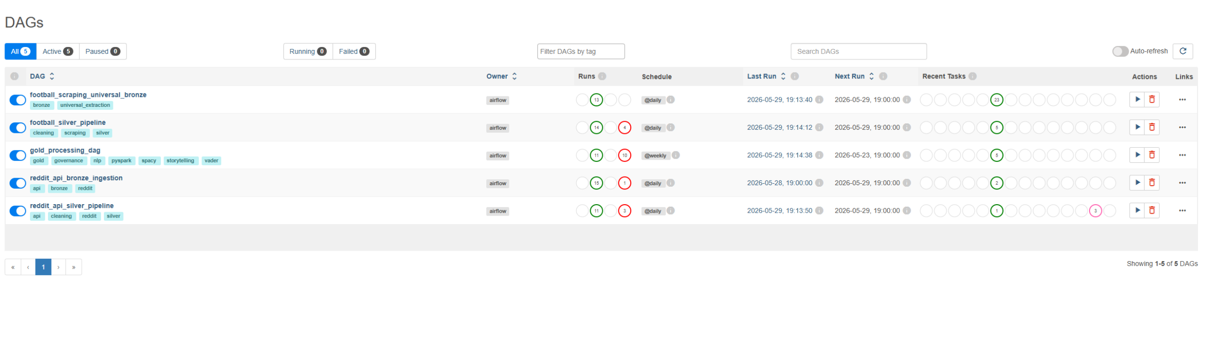

Figure 1. Apache Airflow interface showing the execution history of the Bronze, Silver, and Gold DAGs.

7. Governance Dashboard — Full-Page Screenshot and Component Walk-through

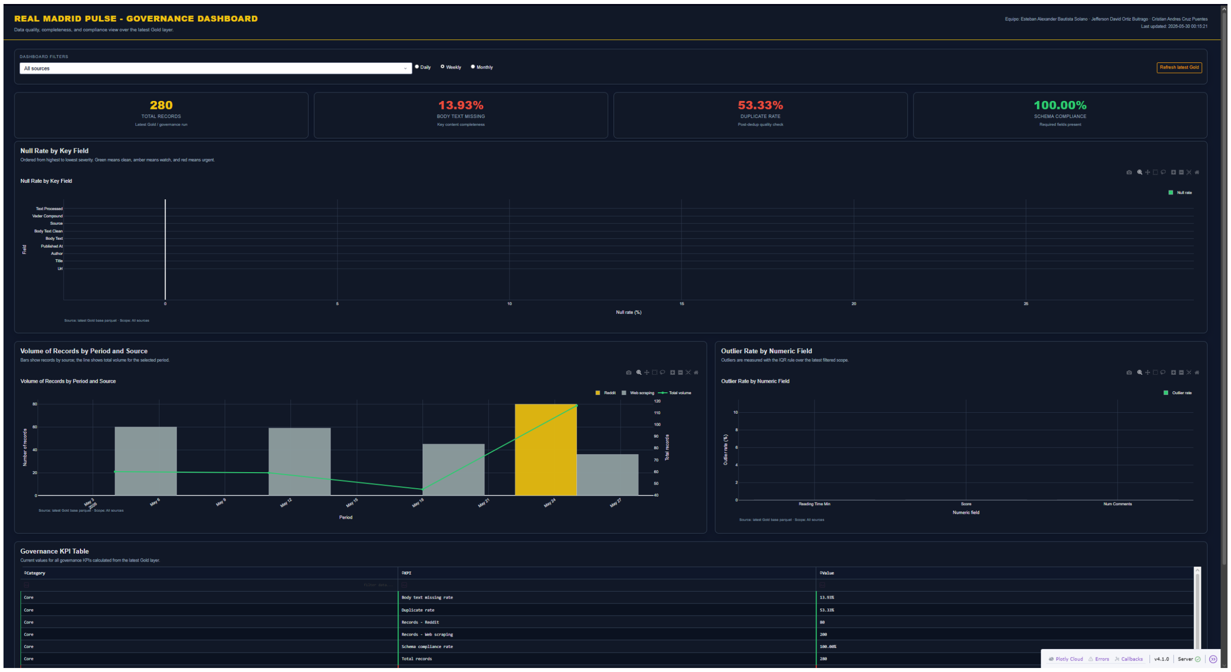

Figure 2. Governance Dashboard — full view at localhost:8052, populated with real Gold layer data from 280 ingested records across Reddit and Football-España sources.

The governance dashboard presents the operational health of the Real Madrid Pulse pipeline through a layered monitoring interface. The following subsections describe each visible component and the KPI it communicates.

Header and Timestamp

The dashboard header displays the project title, team identification, and the timestamp of the most recently available Gold Parquet file. This timestamp provides immediate confirmation that the pipeline has executed and that the displayed metrics reflect the current state of the data lake. In the screenshot, the timestamp corresponds to the latest successful run of the `gold_processing_dag`.

Filter Bar

Figure 3. Filter bar showing source dropdown, period selector, and refresh button.

The filter bar contains three interactive controls that dynamically recalculate all dashboard metrics without reloading the page:

- **Source dropdown** (*All Sources / Reddit / Web Scraping*): filters the Gold base DataFrame by the `source` field. When a specific source is selected, all KPI cards, charts, and the governance table are recalculated exclusively over that subset.
- **Period selector** (*Daily / Weekly / Monthly*): controls the temporal aggregation level of the Volume Over Time chart. Daily granularity reveals day-level ingestion patterns; weekly and monthly views summarize longer trends.
- **Refresh Latest Gold** button: forces an immediate reload of the most recent Gold Parquet files from `datalake_gold/`, bypassing the five-minute auto-refresh interval.

KPI Summary Cards

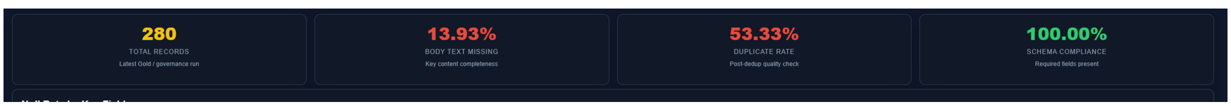

Figure 4. KPI summary cards showing total records, body text missing rate, duplicate rate, and schema compliance.

The four KPI cards provide an immediate overview of the most critical governance indicators:

1. **Total Records (280)**: confirms the number of unique records present in the latest Gold Parquet file after global deduplication. This count includes 200 articles from Football-España and 80 posts from Reddit.
2. **Body Text Missing (13.93%)**: indicates the percentage of records where the `body_text` field is empty or absent. This rate is expected for Reddit, where title-only posts are common. The VADER sentiment analysis uses `title` as a fallback for these records.
3. **Duplicate Rate (53.33%)**: reflects the proportion of records that were identified as duplicates across ingestion runs and removed during the global deduplication step in Task 1 of the Gold DAG. This high rate is expected because the Reddit Bronze DAG ingests the same recent posts across multiple daily runs; the Silver and Gold deduplication layers ensure only unique records reach the analytical layer.
4. **Schema Compliance (100%)**: confirms that all 280 records contain values for all required fields (`url`, `title`, `author`, `published_at`, `body_text`, `source`). A compliance rate of 100% validates the schema enforcement logic applied in both Silver DAGs.

Null Rate by Key Field

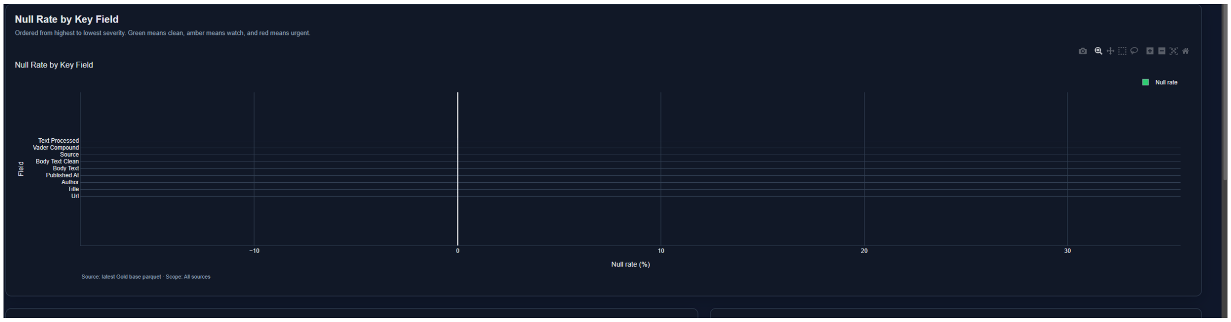

Figure 5. Null rate chart showing 0% missing values across all key fields after Silver preprocessing.

The null rate chart ranks key fields by their percentage of missing values. All fields return a null rate of 0%, confirming that the Silver preprocessing pipeline successfully handled all missing values through the following strategies: sentinel imputation (`author` → *Unknown*), boolean flagging (`body_text_missing`), median imputation (`reading_time_min`), and date fallback (`published_at` → `published_time`). The absence of null values at the Gold layer validates that the completeness strategy defined in Workshop 2 was correctly implemented and carried through to the final analytical layer.

Volume of Records by Period and Source

Figure 6. Volume chart showing weekly ingestion activity by source, with total volume trend line.

The volume chart displays the number of records ingested per period, grouped by source. The grouped bars allow comparison of Reddit and Football-España contribution in each period; the overlaid line shows total volume. The chart reveals that the week of May 24 concentrates the highest volume, corresponding to the first full execution of the Reddit Bronze DAG with pagination enabled (up to 500 posts per run). Earlier weeks show only scraping activity, consistent with the ingestion timeline of the project.

Outlier Rate by Numeric Field

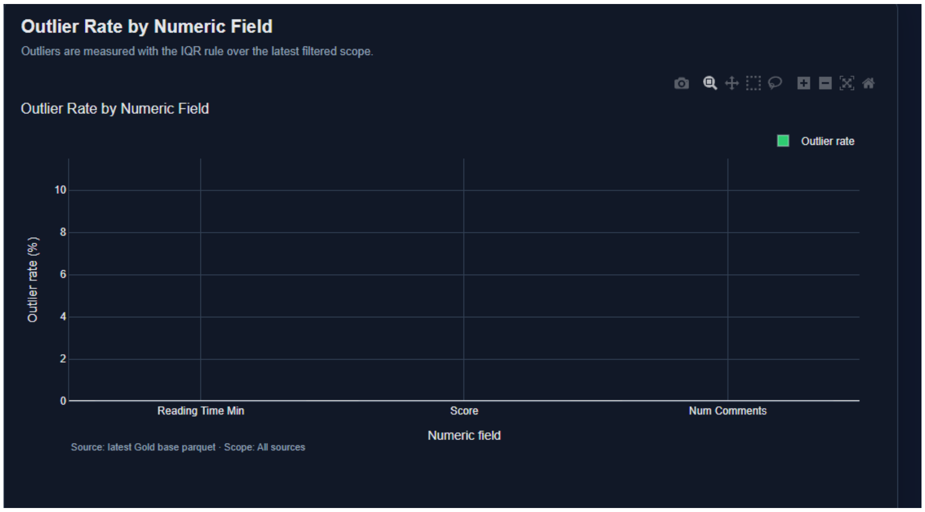

Figure 7. Outlier rate chart showing 0% outliers across all numeric fields after IQR capping in Silver DAGs.

The outlier rate chart shows the percentage of records whose numeric values fell outside the IQR bounds before capping. All three fields (`reading_time_min`, `score`, `num_comments`) return an outlier rate of 0% in the Gold layer. This result is expected: the IQR-based capping applied in both Silver DAGs clips extreme values before saving the Parquet files, so by the time data reaches Gold, no record falls outside the computed bounds. This confirms that the outlier treatment pipeline is functioning correctly end-to-end.

Governance KPI Table

Governance KPI Table
Current values for all governance KPIs calculated from the latest Gold layer.

Category	KPI	Value
Core	Body text missing rate	13.93%
Core	Duplicate rate	53.33%
Core	Records - Reddit	80
Core	Records - Web scraping	200
Core	Schema compliance rate	100.00%
Core	Total records	280
Completeness	Null rate - author	0.00%
Completeness	Null rate - body text	0.00%
Completeness	Null rate - body text clean	0.00%
Completeness	Null rate - published at	0.00%
Completeness	Null rate - source	0.00%
Completeness	Null rate - text processed	0.00%
Completeness	Null rate - title	0.00%

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Plotly Cloud Errors Callbacks v4.1.0 Server

Figure 8. Governance KPI table showing all calculated metrics grouped by category with sorting and filtering support.

The governance KPI table presents all computed metrics organized into five categories: *Core*, *Completeness*, *Quality*, *Ingestion*, and *Distribution*. The table supports native sorting and filtering, allowing technical users to drill into specific metric groups. Color-coded left borders distinguish categories visually: green for Core metrics, red for Completeness issues, yellow for Quality indicators, and gray for Ingestion compliance. The table serves as the detailed audit layer of the dashboard, complementing the high-level charts above with exact numeric values for all governance KPIs computed in Task 4 of the Gold DAG.

7.1 Storytelling Dashboard — Full-Page Screenshot

Figure 9 presents the main landing page of the Storytelling Dashboard. This section provides an executive summary of the sentiment analysis results by combining key performance indicators, sentiment distribution metrics, and temporal trend visualizations.

At the top of the dashboard, users can identify the active data sources, the latest update timestamp, and the overall sentiment score calculated from the integrated Gold dataset. The KPI cards summarize the total number of analyzed mentions, the dominant sentiment category, the global sentiment score, and the number of records contributed by each source. A sentiment gauge and distribution indicators provide an immediate interpretation of the current public perception surrounding Real Madrid.

The lower section focuses on temporal analysis. The sentiment trend chart displays the evolution of average sentiment over time, while the volume chart highlights fluctuations in discussion activity across sources. Together, these visualizations allow users to identify periods of increased attention and determine whether major events were associated with positive or negative shifts in public opinion.

Figure 9. Dashboard interface – Real Madrid Pulse general overview displaying global sentiment metrics, active sources, and temporal evolution of fan sentiment.

7.2 Storytelling Dashboard — Source Comparison and Vocabulary Analysis

Figure 10 illustrates the comparative analysis between Reddit discussions and Football-España news articles. This section was designed to identify differences in sentiment behavior and thematic focus across the two information sources.

The sentiment comparison visualizations reveal how each source contributes to the overall perception of Real Madrid by displaying positive, neutral, and negative distributions separately. This allows users to determine whether fan discussions and media coverage exhibit similar or contrasting emotional patterns.

The dashboard also incorporates keyword and bigram frequency analyses generated from the NLP pipeline. These visualizations highlight the most frequently discussed terms and recurring word combinations after preprocessing and stopword removal. As a result, users can quickly identify the dominant topics driving online conversations and media narratives during the analyzed period.

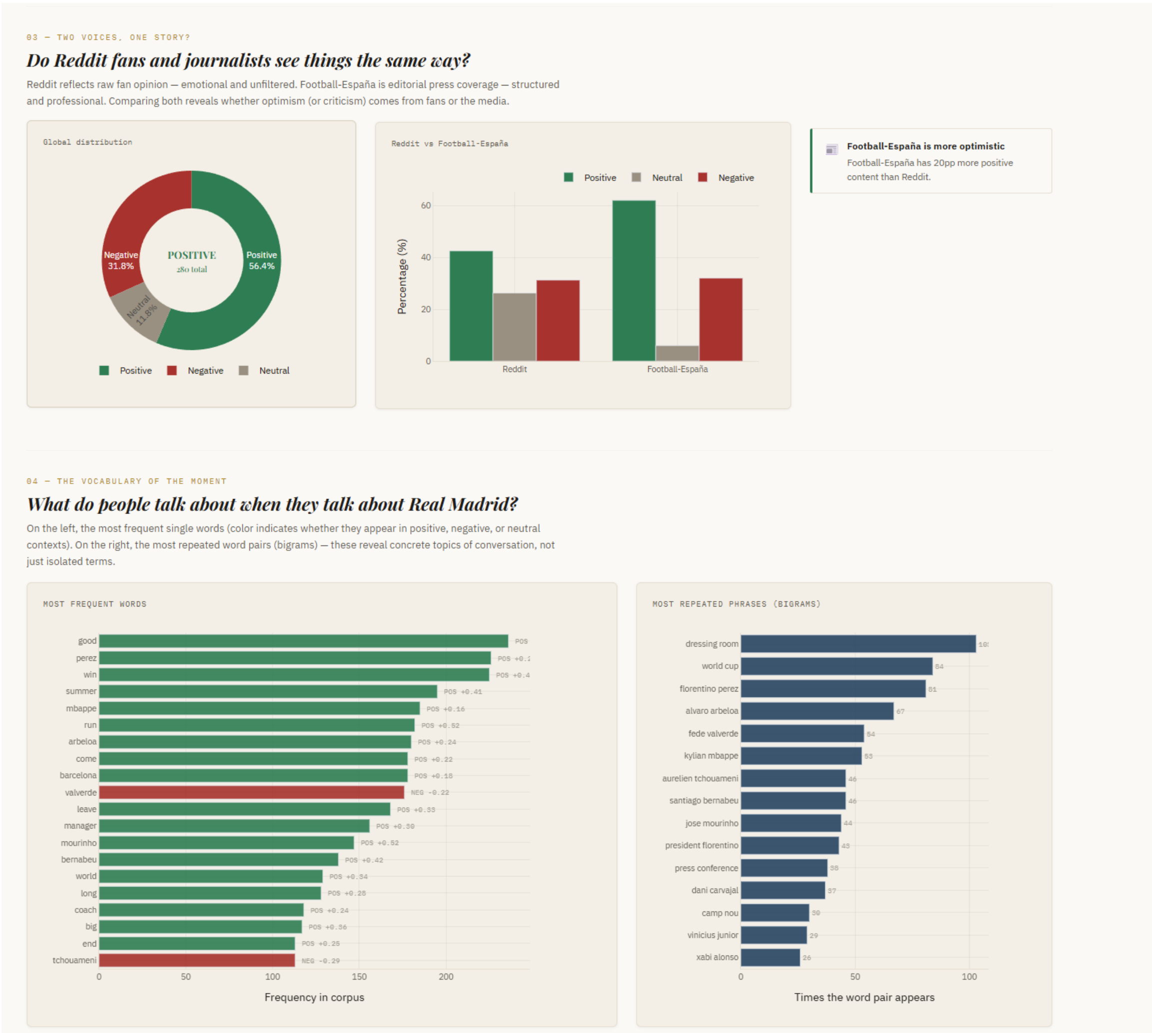

Figure 10. Sentiment and vocabulary comparison – Global distribution contrast between Reddit and Football-España, alongside the most frequent words and bigrams.

7.3 Storytelling Dashboard — Entity Sentiment Analysis

Figure 11 presents the entity-level sentiment analysis module. This section aggregates sentiment scores around key protagonists and topics frequently mentioned within the dataset, including players, coaches, rivals, and transfer-related discussions.

The entity ranking visualization orders aspects according to their average sentiment score and mention frequency, making it possible to identify which subjects generate the most positive or negative reactions among supporters and journalists. Complementary annotations highlight notable findings and provide contextual explanations for extreme sentiment values.

This component transforms the overall sentiment analysis into a more interpretable storytelling experience by linking sentiment directly to real-world football entities and events.

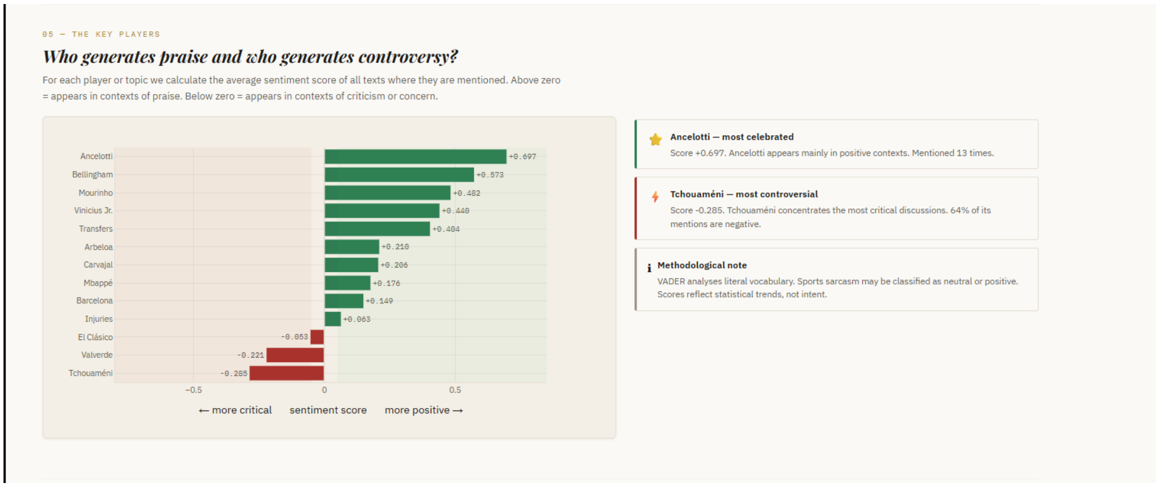

Figure 11. Entities sentiment index – Ranking of protagonists based on average mood scores, highlights of praise, and methodological remarks on VADER lexicon analysis.

7.4 Storytelling Dashboard — Protagonist Deep Dive

Figure 12 shows the detailed exploration module for individual protagonists. After selecting a specific entity, the dashboard provides a focused analysis of the language and sentiment associated with that subject.

The keyword frequency charts separate the most common positive and negative terms that co-occur with the selected protagonist, helping users understand the factors that contribute to favorable or unfavorable sentiment. Additionally, representative text excerpts extracted from Reddit posts and news articles are displayed to provide qualitative evidence supporting the numerical sentiment indicators.

This module bridges quantitative sentiment metrics and qualitative textual interpretation, allowing users to understand not only the sentiment score itself but also the underlying reasons behind public perception.

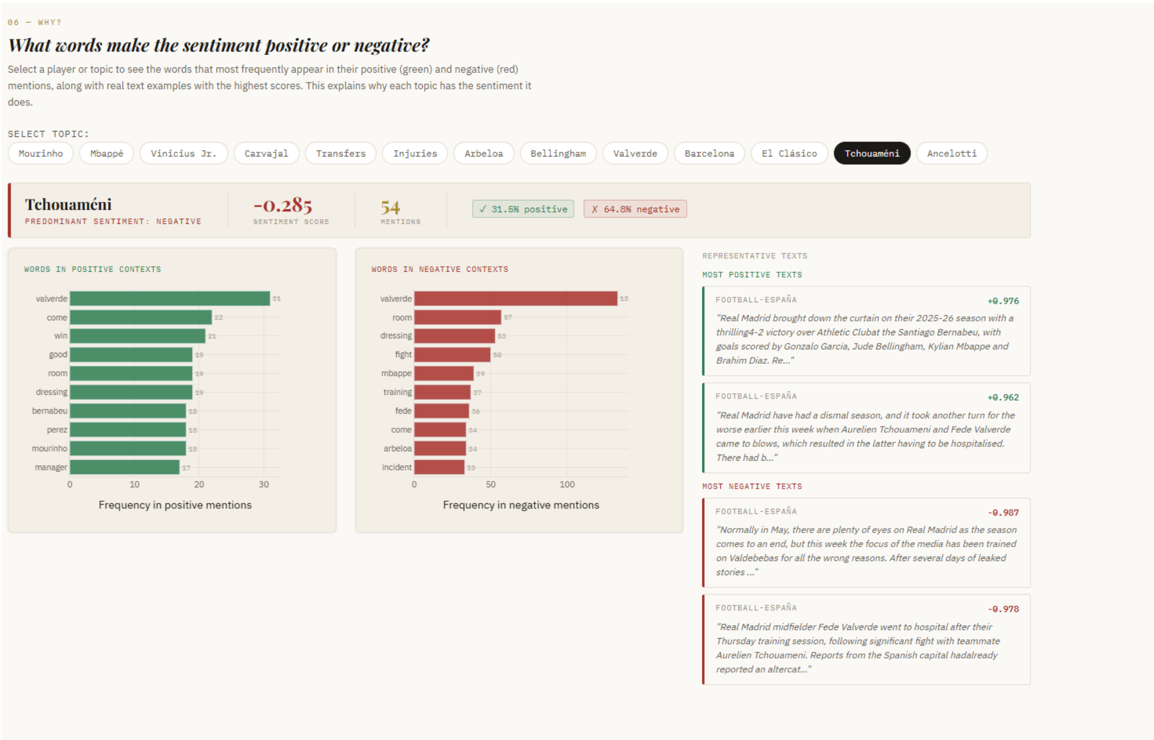

Figure 12. Protagonist deep dive – Specific breakdown of keyword frequencies in positive and negative contexts, paired with highly representative source texts for Bellingham.

8. Reproducibility

8.1 Startup Sequence

The entire project is reproducible from a fresh repository clone using the following command sequence:

```
# 1. Start Docker Compose environment
docker-compose up --build -d

# 2. Access Airflow UI at localhost:8080 and trigger DAGs in order,
#    or use the CLI:
docker-compose exec airflow-webserver \
    airflow dags trigger football_bronze_pipeline
docker-compose exec airflow-webserver \
    airflow dags trigger reddit_bronze_pipeline
docker-compose exec airflow-webserver \
    airflow dags trigger football_silver_pipeline
docker-compose exec airflow-webserver \
    airflow dags trigger reddit_api_silver_pipeline
docker-compose exec airflow-webserver \
```

```
    airflow dags trigger gold_processing_dag

# 3. Launch dashboards (each in a separate terminal)
python dashboard/governance_app.py    # accessible at localhost:8050
python dashboard/storytelling_app.py  # accessible at localhost:8051
```

Listing 3. Full startup and execution sequence

8.2 Repository Structure

The final repository is organized following the Medallion Architecture layer separation, with dedicated folders for orchestration, dashboards, notebooks, and workshop deliverables. The structure below reflects the state of the repository at the `v1.0-final` tag.

```
prog_analisis_datos/
├── datalake_bronze/ ...................................... Raw JSON files per ingestion run
│   ├── reddit_api_realmadrid_YYYYMMDD_HHMMSS.json
│   └── web_scraping_realmadrid_YYYYMMDD_HHMMSS.json
├── datalake_silver/ ...................................... Processed Parquet files
│   ├── reddit_api_realmadrid_YYYYMMDD_HHMMSS.parquet
│   ├── web_scraping_realmadrid_YYYYMMDD_HHMMSS.parquet
│   └── outlier_reports/ ...................................... Boxplots and metrics per run
├── datalake_gold/ ...................................... Analytical summaries
│   ├── gold_realmadrid_YYYYMMDD_HHMMSS.parquet
│   ├── governance_YYYYMMDD_HHMMSS.parquet
│   ├── sentiment_distribution_YYYYMMDD_HHMMSS.parquet
│   ├── sentiment_trend_YYYYMMDD_HHMMSS.parquet
│   ├── top_keywords_YYYYMMDD_HHMMSS.parquet
│   ├── keyword_sentiment_YYYYMMDD_HHMMSS.parquet
│   ├── source_comparison_YYYYMMDD_HHMMSS.parquet
│   ├── volume_trend_YYYYMMDD_HHMMSS.parquet
│   ├── aspect_sentiment_YYYYMMDD_HHMMSS.parquet
│   └── storytelling_summary_YYYYMMDD_HHMMSS.parquet
├── airflow/
│   └── dags/ ...................................... All Airflow DAG definitions
│       ├── reddit_api_bronze_ingestion_dag.py
│       ├── football_scraping_universal_bronze.py
│       ├── reddit_api_silver_processing_dag.py
│       ├── silver_processing_web_dag.py
│       └── gold_processing_dag.py
├── dashboard/ ...................................... Plotly Dash applications
│   ├── storytelling_dashboard.py ...................................... localhost:8051
│   └── governance_dashboard.py ...................................... localhost:8052
├── notebooks/ ...................................... Exploratory analysis
│   ├── notebook_1_data_quality_findings.ipynb
│   └── notebook_2_gold_preview.ipynb
├── workshops/ ...................................... Workshop deliverables
│   ├── workshop_1/
│   ├── workshop_2/
│   ├── workshop_3/
│   └── workshop_4/
├── Dockerfile
├── docker-compose.yml
├── requirements.txt
└── README.md
```

The repository follows these organizational principles:

- **Data lake layers** (`datalake_bronze/`, `datalake_silver/`, `datalake_gold/`) are mounted as Docker bind-mount volumes, ensuring that all pipeline outputs persist on the host filesystem across container restarts. Timestamp-based file naming provides natural chronological ordering and full traceability from Bronze JSON to Gold Parquet.

- **DAG files** are stored under `airflow/dags/` and mounted into the Airflow scheduler container via Docker volume, enabling live DAG reloading without container rebuilds. Five DAGs cover the complete pipeline: two Bronze ingestion DAGs, two Silver processing DAGs, and one Gold aggregation DAG.
- **Dashboard files** are isolated in `dashboard/` and run as independent Plotly Dash applications on ports 8051 (storytelling) and 8052 (governance). Both dashboards read exclusively from `datalake_gold/` and never connect directly to Bronze, Silver, or PostgreSQL.
- **Notebooks** in `notebooks/` document the data quality findings and the Gold layer preview, fulfilling the Workshop 3 exploratory analysis requirement.
- **Workshop deliverables** are organized in numbered subfolders under `workshops/`, with each folder containing the corresponding PDF report.

To reproduce the complete pipeline from a fresh clone, follow the instructions in `README.md`: build the Docker image, start the services with `docker compose up`, trigger the Bronze DAGs, wait for Silver processing, trigger the Gold DAG, and launch the dashboards.

9. Challenges and Resolutions

9.1 Cold-Start Empty State

Challenge: When the dashboard is launched before any Gold DAG run has completed, all `load_latest_parquet()` calls return empty DataFrames, causing callbacks to raise exceptions on column access.

Resolution: Every callback begins with an `if not df.empty` guard before accessing any column. If the DataFrame is empty, the callback returns an empty `go.Figure()` with the base layout applied. KPI cards display `"-"` as a value placeholder. The narrative card shows `"No data available. Run the Gold DAG first."` This ensures the dashboard is usable and informative in a cold-start state without crashing.

9.2 Daily vs. Weekly Granularity Toggle

Challenge: The Gold DAG produces pre-aggregated trend files at weekly granularity only. Supporting a daily view without modifying the Gold DAG required an alternative approach.

Resolution: The `update_trend_charts` callback reads the full Gold base dataset and computes the groupby aggregation at runtime based on the toggle value (`"day"` or `"week"`). The `published_at` column is parsed to datetime and the `day` and `week` columns are derived by `load_full_gold()`, keeping both granularities available without any Gold DAG changes.

9.3 Keyword Sentiment Color Merge

Challenge: The `top_keywords` Parquet file contains frequency data only, while sentiment mode is stored in a separate `keyword_sentiment` file. A column lookup by keyword string was needed to assign a color to each bar.

Resolution: The `update_keywords` callback loads both files and performs a `pd.merge(..., on="keyword", how="left")` to enrich the top-15 keywords with their sentiment mode. Keywords not found in the sentiment file have `sentiment_mode` filled with `"neutral"`, ensuring every bar receives a valid color without raising a key error.

9.4 Aspect Source Filtering

Challenge: The `aspect_sentiment` Parquet file contains rows for each aspect at three levels (`source == "all"`, `"reddit"`, `"scraping"`). Plotting all rows without filtering would produce three bars per aspect and clutter the chart.

Resolution: The `update_aspect_sentiment` callback filters to `df[df["source"] == "all"]` before rendering, displaying only the global compound score per aspect. A fallback uses all rows if no "all" rows are found, to handle edge cases where the Gold aggregation schema differs between DAG runs.

10. Conclusions

Workshop 4 completes the Real Madrid Pulse project by delivering the user-facing presentation layer on top of the Medallion Architecture pipeline built across the previous workshops. The two Plotly Dash dashboards fulfill the architectural separation between technical governance monitoring and functional user storytelling, each reading exclusively from Gold Parquet files as the single source of truth.

Analytical Findings

The central finding of the project is that public discourse about Real Madrid during the analyzed period (May 2026) is predominantly positive. Of the 280 records processed from two independent sources, 56.43% were classified as positive sentiment, with a global VADER compound score of +0.207, confirming an overall optimistic tone in both fan discussion and sports media coverage.

At the entity level, Ancelotti and Bellingham emerged as the most positively discussed figures, appearing consistently in contexts of praise, tactical recognition, and performance appreciation. At the opposite end, Tchouaméni concentrated the most critical discourse, driven largely by Football-España coverage of the Tchouaméni–Valverde training incident, which produced several highly negative articles that pulled the compound score significantly below the neutral threshold.

The source comparison analysis revealed a structural difference between the two ingestion channels: Football-España produced 20 percentage points more positive content than Reddit, consistent with the contrast between structured sports journalism and unfiltered fan discussion. Reddit showed substantially higher neutrality rates, reflecting the conversational and question-driven nature of community posts. This finding validates the design decision to maintain two separate sources — the contrast between them is itself an analytical result, not a data quality problem.

Keyword analysis of the NLP-processed corpus identified *week*, *good*, *win*, and *season* as the most frequent terms, with player names (Mbappé, Arbeloa, Bellingham, Valverde) appearing consistently across both sources, confirming that personnel topics dominated the discourse during the analyzed window.

Technical Conclusions

The complete pipeline from Bronze ingestion to Gold aggregation executed successfully end-to-end, producing traceable, timestamped outputs at each layer with 100% schema compliance and 0% null rates across all key fields at the Gold layer. The governance dashboard confirmed that the preprocessing strategies defined in Workshop 2 — sentinel imputation, IQR capping, boolean flagging, and date fallback — were correctly applied and produced a clean analytical dataset.

The key technical limitation identified is that VADER, as a lexicon-based model, cannot detect sarcasm, irony, or implicit criticism common in football fan discourse. A post such as *"Mbappé invisible again, just elite"* would be classified as positive by VADER due to the word *elite*, despite the critical intent. Sentiment scores should therefore be interpreted as statistical tendencies over large corpora rather than precise classifications of individual texts. A transformer-based model fine-tuned on sports commentary is the recommended upgrade for future iterations.

11. Lessons Learned

Schema Consistency Must Be Enforced at Write Time

The most consequential technical lesson was that Parquet schema consistency cannot be assumed across ingestion runs — it must be enforced explicitly at the point of writing. The `score` field in the Reddit Silver DAG was saved as `BIGINT` in early runs and as `DOUBLE` in later ones, because `pd.read_json()` infers types from values and integer-looking floats (e.g., `5.0`) are cast to `int64` when all values happen to be whole numbers. This caused the PySpark UNION in the Gold DAG to fail with a schema mismatch error. The resolution — adding `.astype(float)` before every `to_parquet()` call — is trivial, but finding the root cause required inspecting individual Parquet schemas with PySpark `printSchema()`. In production pipelines, schema registries such as Apache Avro or Delta Lake solve this class of problem structurally.

Layer Responsibility Boundaries Improve Pipeline Quality

Relocating spaCy lemmatization from the Silver DAG to the Gold DAG during Workshop 3 was initially a refactoring decision, but it proved to be architecturally correct. The Silver layer's responsibility is cleaning and normalization; the Gold layer's responsibility is analytical enrichment. Mixing NLP processing into Silver made the Silver DAG slower, harder to debug, and tightly coupled to a specific NLP library. After the separation, Silver ran faster, its outputs were more reusable, and the Gold NLP pipeline could be modified independently without affecting Silver Parquet files.

Real Data Always Reveals Unanticipated Problems

Every preprocessing strategy defined in Workshop 2 was based on assumptions about the data. Running the pipeline against real data in Workshops 3 and 4 revealed problems that could not have been anticipated: the `reading_time_min` field returning a constant value of 2.0 for all scraped articles (indicating a site-level default rather than per-article metadata), Reddit returning multilingual posts despite an English-focused query (requiring the `langdetect` filter in the Bronze DAG), and the `Variable.set()` Airflow call failing on second execution due to a uniqueness constraint that requires a `delete` before `set`. These findings could only be discovered by running the pipeline, not by designing it. Experimentation and iteration are not signs of incomplete planning — they are the primary method of data engineering.

12. Future Work

- **Transformer-based sentiment model.** Replace VADER with a RoBERTa or BERTweet model fine-tuned on football tweets or sports commentary. This would improve detection of sarcasm, implicit criticism, and domain-specific sentiment expressions that VADER systematically misclassifies.
- **Expanded source coverage.** Add additional Bronze ingestion DAGs for Twitter/X (via the Academic Research API), BBC Sport, and Marca to increase corpus diversity, reduce source bias, and enable cross-language sentiment comparison between English and Spanish coverage.
- **Incremental Gold processing.** Modify the Gold DAG to append new records to existing Gold Parquet files rather than reprocessing the full Silver layer on each run. This would reduce Gold DAG execution time as the corpus grows and is the standard pattern in production data lakes.
- **Event-driven filtering in dashboards.** Add a date range filter to the storytelling dashboard to enable match-specific analysis — for example, filtering sentiment to the 48-hour window around a specific Champions League match to measure fan reaction to the result.